

Q SGD Momentum Analysis

Our main refactoring step for this notebook was the addition of a main helper function for the redundant plotting code. Previously, this notebook contained a series of fairly redundant cells with repeated plotting code that compared gradients, parameters, and losses. We wrote a function **plot_optimization_comparison**, which handles comparisons between multiple methods of descent, as well as the different possible metrics with which to compare methods. This function also has toggleable options to perform transformations, such as applying a log scale to the y-axis. This will enable students to plot comparisons between various methods of descent in a streamlined process, and gives the reader a better idea of what, specifically, each plotting code block is meant to demonstrate.

We additionally added typehints to all implemented functions, adhering to Python Enhancement Proposal 484, and we also added docstrings to all functions. These refactoring steps enhanced the code's readability, and the typehints allow for further verification that relevant functions are being used correctly. For our docstrings, we provided our functions, as well as a brief description of the purposes of our functions, and prompted Claude to give us a properly formatted NumPy-style docstrings to add to our functions, with full descriptions of our functions' input and output values, as well as usage examples.